

FAST – NUCES



Object Oriented Programming Lab Manual

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Lab 5 – OOP Constructors/ Destructor and Member Functions

Objectives

- Constructors (Default/ overloaded/ copy)
 - Destructor
 - Member functions
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Read carefully and attempt following task

“HeroMissionDate” (Powerpuff Girls Edition)

Create a class named **HeroMissionDate** that represents important mission dates in the city of Townsville.

Private Data Members

The class should have the following private data members:

- Day – The day of the mission
- Month – The month of the mission
- Year – The year the mission took place

Constructors

1. Default Constructor

Initialize the object to:

 **January 1, 1926**

(The day Professor Utonium started his laboratory journal)

2. Overloaded Constructor

Create another constructor that takes:

- day
- month
- year

This allows Blossom, Bubbles, or Buttercup to set a specific mission date.

Destructor

Write a destructor that prints:

"Saving Townsville ends... Destructor called!"

This message appears when a HeroMissionDate object is destroyed.

Member Functions

1. Print()

Print the mission date in the format:

dd/mm/yyyy

Example:

14/09/2026

2. Input()

Allow the user to enter:

- Day
- Month
- Year

To set a new Townsville mission date.

3. SetYear()

Create a function that modifies the mission year.

4. GetDay() and GetMonth()

Functions that return:

- The mission day
- The mission month

Object Passing & Returning

1. Compare(HeroMissionDate otherDate)

Compare two mission dates and return:

- 1 → If the current mission date is later
- -1 → If earlier
- 0 → If both are the same

Example:

If Blossom's mission happened after Buttercup's → return 1.

2. IncrementMonth()

Write a function that:

- Returns a **new HeroMissionDate object**
- Increases the month by one
- Does **not** modify the original object

(Like scheduling the next battle with Mojo Jojo next month)

Main()

Test all your methods in main



-----Happy Coding 😊-----



Submission Instructions

- Submit your tasks on Google Classroom within given deadline
- Submit solution file of each task separately for example “LAB01-TASK01.cpp”, “LAB01-TASK02.cpp” and so on
- Also submit screenshots of output of each task for example “LAB01-TASK01.png”, “LAB01-TASK02.png” and so on
- “.cpp” is the actual file where all your code is saved.
- Always submit .cpp files when asked for submission
- Example path of .cpp file is

Desktop\Your_Roll_No\LAB01-TASK01\LAB01-TASK01\LAB01-TASK01.cpp

Copy your files from this path

